Southwire®
POWER GENERATION SOLUTIONS

SK-3X1 POWER CAMPUS

# STRATEGY MONTHLY UPDATE

## EVERY MEGAWATT TRAVELS THROUGH A CABLE




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# ONE CAMPUS.
# EVERY CABLE.

A 3x1 combined-cycle plant with RICE engines, carbon capture, battery storage, a modular power yard, an LNG terminal and a behind-the-meter data centre: one campus, and from the 230 kV grid connection to the last instrument loop, all of it runs on cable. The next slides follow that cable through the plant.


| | | |
|---|---|---|
| 01 | GRID CONNECTION | SLIDE 04 |
| 02 | POWER ISLAND · TURBINE HALL | SLIDE 06 |
| 03 | CARBON CAPTURE | SLIDE 09 |
| 04 | BATTERY STORAGE | SLIDE 10 |
| 05 | MODULAR & TEMPORARY POWER | SLIDE 11 |
| 06 | INSTALLED FASTER | SLIDE 12 |




THE PLANT AT A GLANCE

# WHERE THE CABLE GOES.

Take the steel away and this is what is left: one network, from the 230 kV grid connection to the last instrument loop, in every zone of the campus.

## BY THE NUMBERS

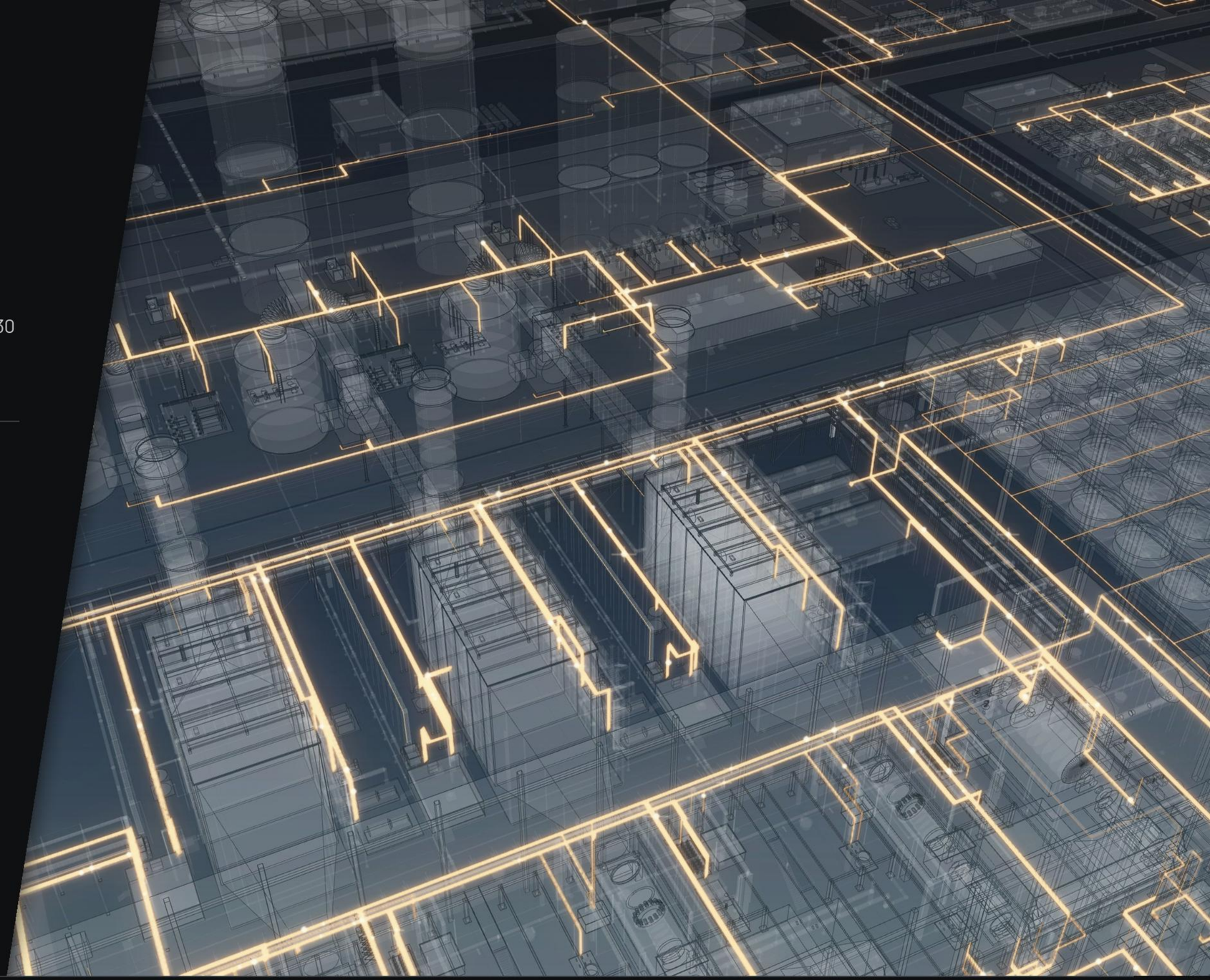
**4.6 MI**
of cable tray routes on the racks and in the halls

**4.9 MI**
of duct-bank and buried cable routes

**4,946 FT**
of 230 kV underground cable routes

**418**
pieces of equipment with their own cable applications

Route lengths measured along the modelled tray, duct-bank and cable routes (not cable footage); data-centre campus excluded. Picture: the 3D model in cable X-ray.



# 230 kV
# TO THE GRID

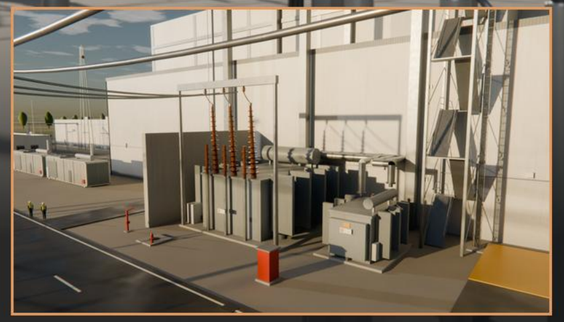
The plant's output leaves through four generator step-up transformers and a breaker-and-a-half 230 kV switchyard. Where the line runs underground, high-voltage XLPE cable rises on cleated risers to sealing ends and surge arresters.

**01 HV XLPE TRANSMISSION CABLE**
Copper conductor, XLPE insulation, aluminium laminate moisture barrier and HDPE jacket for the buried 230 kV run.

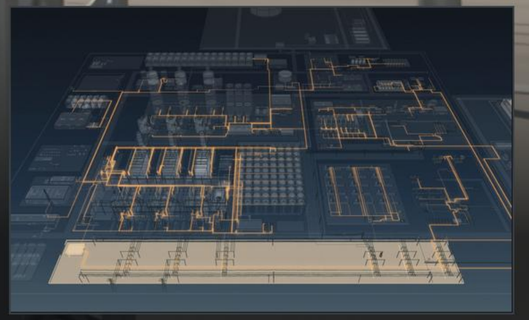
**02 FEWER JOINTS**
Long continuous lengths keep splices out of the duct bank.

**03 ONE SUPPLIER**
Cable, accessories and field support for the complete circuit.

230 kV | 2500 KCMIL CU | XLPE | AL LAMINATE | ICEA S-108-720 | 4/0 BARE CU GROUND



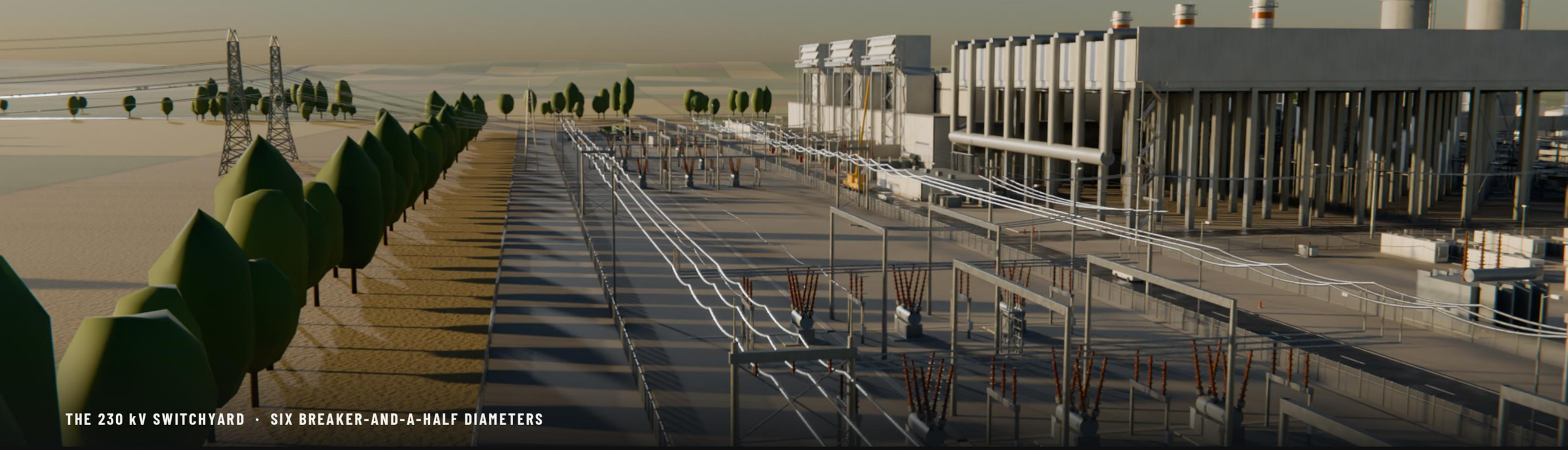
Generator step-up transformers facing the switchyard



WHERE IN THE PLANT



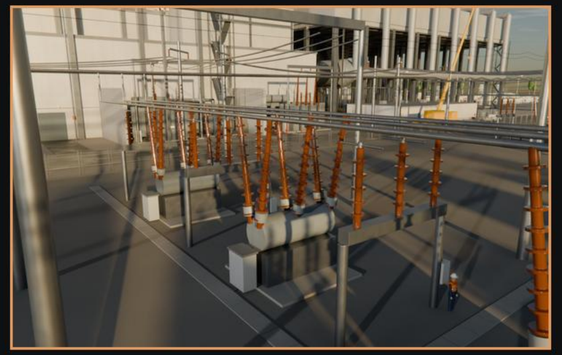
Product data typical; confirm against current Southwire specifications.



THE 230 kV SWITCHYARD · SIX BREAKER-AND-A-HALF DIAMETERS



Along the switchyard at eye level



Bay D2: breakers, disconnects, trench to the relay house

## WHAT RUNS AT THE GRID CONNECTION

### HV XLPE 230 kV

**1/C 2500 KCMIL CU · XLPE · CU WIRE SHIELD · AL LAMINATE · HDPE**

Underground 230 kV circuits and the CCS / modular-yard ties

### BARE COPPER

**4/0 AWG GROUND GRID · RODS · RISERS**

Station ground grid under the yard and every structure

### TYPE TC-ER CONTROL

**600 V MULTICONDUCTOR · EXPOSED-RUN RATED**

Breaker and relay circuits in the yard trenches

### MV-105 35 kV

**1/C 500 KCMIL AL · EPR 133% · 1/3 CN · LLDPE**

Station service and collector feeders

# ARMOR-X
# IN THE TRAY

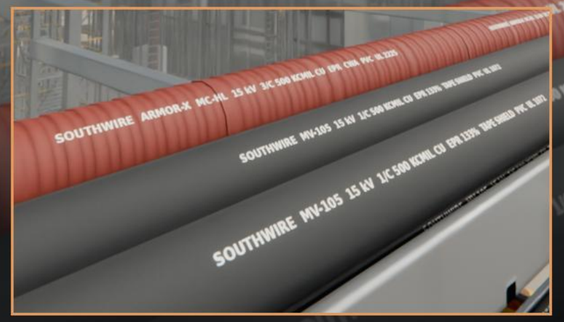
Three H-class gas turbines and a steam turbine share one hall. 15 kV ARMOR-X MC-HL and MV-105 run from the 13.8 kV switchgear across the tray stack to every auxiliary drive, with 600 V ARMOR-X and Type TC-ER for power and control in the gas-turbine enclosures.

**01 HAZARDOUS-LOCATION RATED**
MC-HL: continuously welded, corrugated aluminium armor under a PVC jacket, for the Class I, Div. 2 areas around the gas turbines.

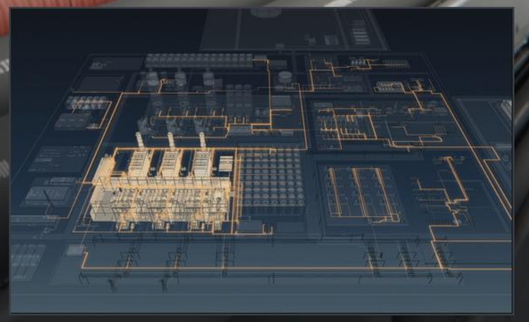
**02 NO CONDUIT**
Armored cable goes straight into the tray: fewer trades, faster pulls.

**03 READABLE EVERYWHERE**
Printed legend on the jacket for identification at every drop.

15 kV MV-105 133% · RED PVC | 600 V CSA RA90-HL · BLACK PVC | UL 1569 / UL 2225 | TYPE TC-ER | MV-105 · UL 1072



The legend on every jacket: ARMOR-X MC-HL 15 kV beside MV-105


WHERE IN THE PLANT

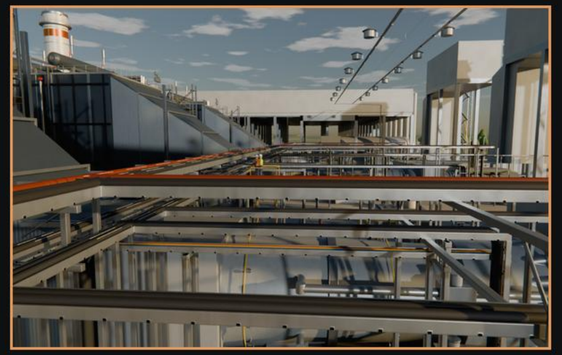


Product data typical; confirm against current Southwire specifications.

OUTAGE PULL · NEW 13.8 kV FEEDER FROM THE LAYDOWN BAY UP INTO THE TRAY



ARMOR-X and MV-105 in the EL 48 tray over GT1



Tray stack along the hall north wall, drops to the GT skids

## WHAT RUNS IN THE TURBINE HALL

### ARMOR-X® MC-HL 15 kV

**500 KCMIL 3/C CU · NL-EPR 25% TAPE SHIELD · GROUND · RED PVC**

MV feeders to auxiliary drives through the gas-turbine areas

### ARMOR-X® MC-HL 600 V

**#14 AWG TO 750 KCMIL · XHHW-2 · CSA RA90-HL · BLACK PVC**

Motor and power circuits on the enclosures and skids

### MV-105 15 kV 133%

**1/C 500 KCMIL CU · EPR · TAPE SHIELD · PVC · UL 1072**

Switchgear to transformers and large motors

### TYPE TC-ER

**600 V POWER AND CONTROL · EXPOSED-RUN RATED**

Control, protection and instrument circuits along the tray

# THREE TURBINES. ONE TRAY STACK.

Three H-class gas turbines and the steam turbine share one hall. Every auxiliary drive, enclosure fan, lube-oil skid and protection relay is fed from the tray stack that runs the length of the hall at EL 30 to 48: medium voltage on top, 600 V power below, control and instrumentation at the bottom.

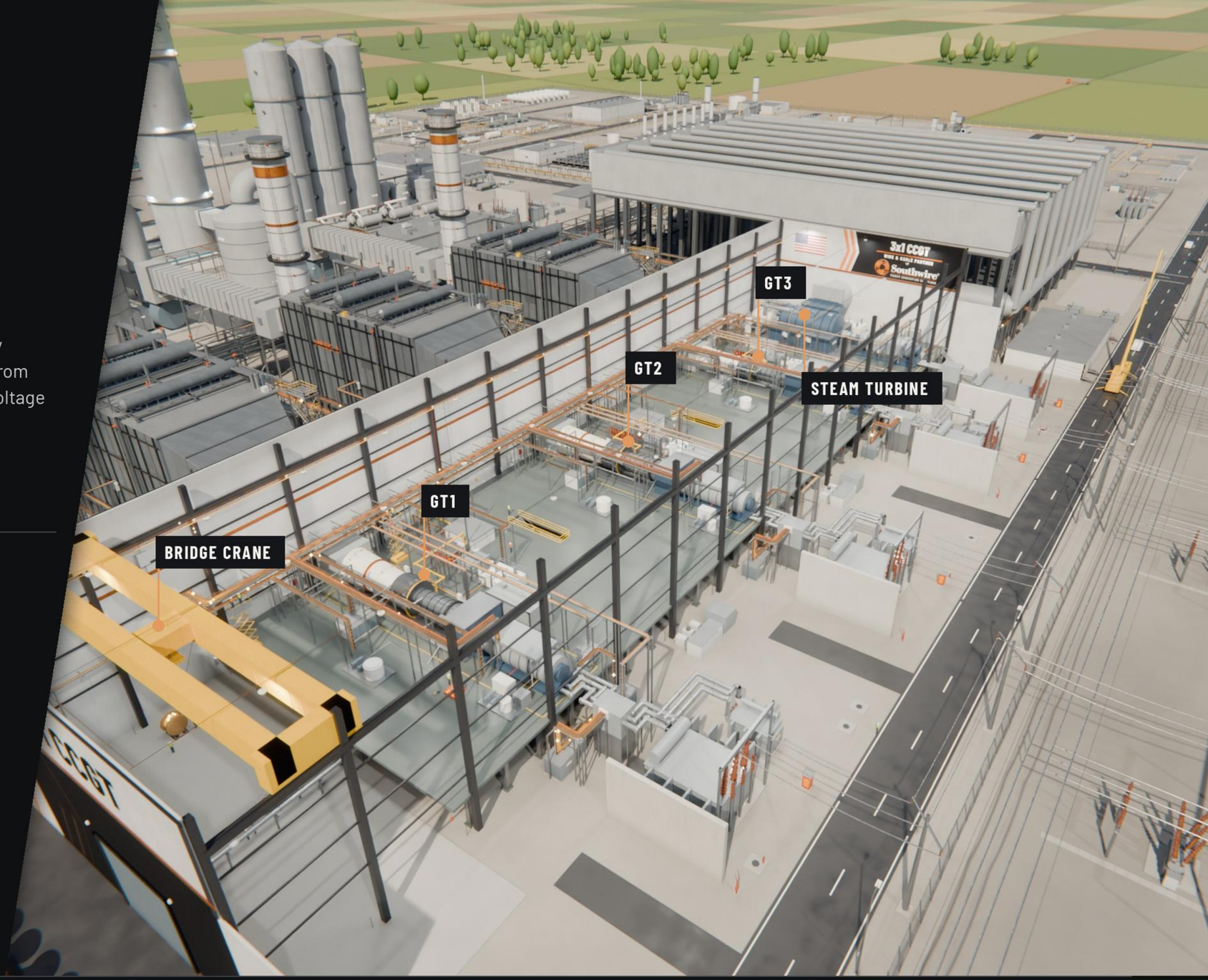
## THE HALL BY THE NUMBERS

**5,604 FT**
of cable tray inside the hall

**1,047 FT**
of it 15 kV medium-voltage tray

**208 FT**
of isolated-phase bus, generators to GSUs

**61**
pieces of equipment in the hall with their own cable






Lengths measured along the modelled routes inside the hall.

# CABLE THAT RUNS CCS

Absorbers, strippers and 16 MW booster fans: post-combustion capture adds a second plant's worth of motors, drives and instruments, all fed from the CCS rack.

**01 ARMOR-X ON THE RACK**
Jacketed armor for the process area's power circuits.

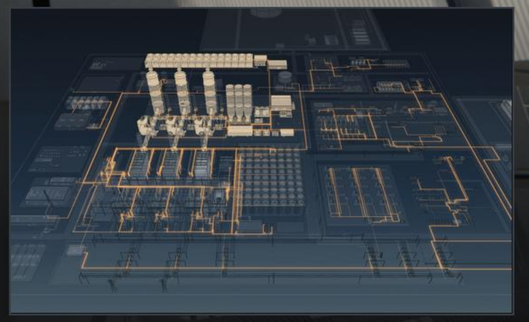
**02 15 KV FEEDERS**
MV-105 to the fans and the CO2 compressors.

**03 INSTRUMENTATION**
PLTC and Type TC-ER for the absorber and regeneration loops.

230 kV XLPE | 15 kV MV-105 | 600 V ARMOR-X | TYPE TC-ER | PLTC



CCS transformers T-1 / T-2: 230 kV terminations and the tray riser



WHERE IN THE PLANT

# DC IN THE TRENCH

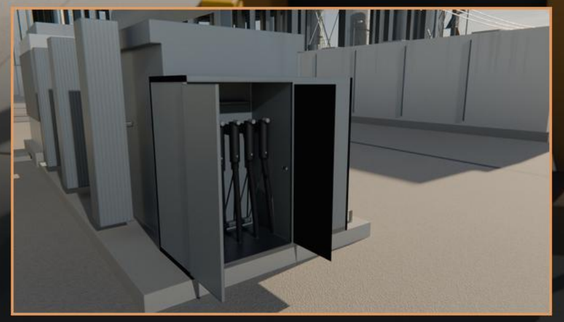
The battery yard pairs containers with power-conversion skids. 1,500 V DC conductors run in precast trenches; 35 kV MV-105 collects the skids to the BESS substation.

**01 PV / RHW-2**
Sunlight- and moisture-resistant 2 kV conductors for the DC side.

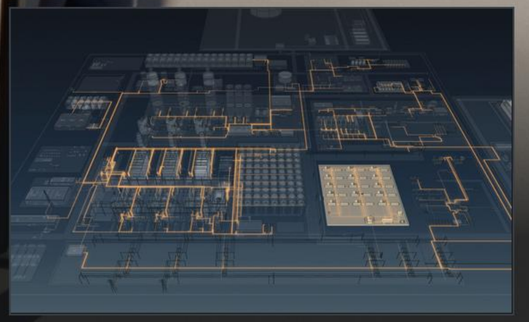
**02 MV COLLECTION**
35 kV MV-105 loop feeds with load-break elbows.

**03 PULLED ON SITE**
Crews pay off from reels straight into the open trench.

2 kV PV / RHW-2 | UL 4703 | 35 kV MV-105 | UL 1072



PCS / MV skid: 34.5 kV elbow terminations to the collector



WHERE IN THE PLANT

# POWER THAT MOVES

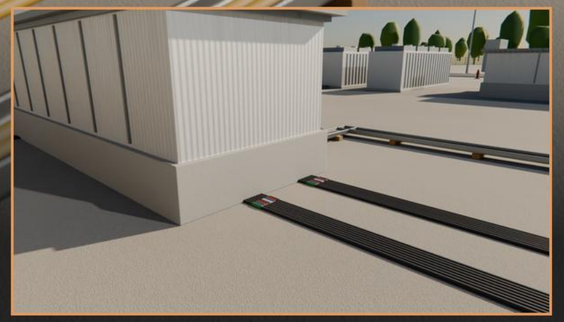
Gensets, trailer turbines and bridge-power fleets come and go. Portable power cable lies on the pad, crosses lanes under drive-over protectors and plugs in with cam-locks; bridge units feed through ground trays to the permanent duct bank.

**01 TYPE SHD-GC**
Flexible 15 kV portable power cable for the MV feeders.

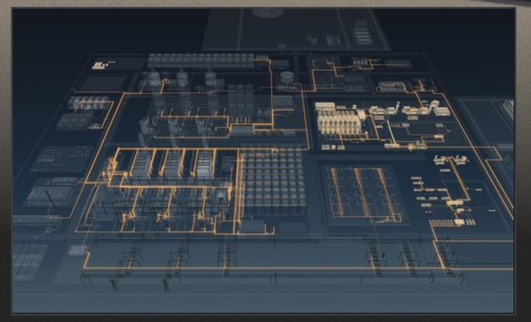
**02 TYPE W**
2000 V single conductors with cam-lock connections for 480 V hookups.

**03 MV-105 LEADS**
Generator leads from the TM2500 terminal boxes in ground trays.

15 kV SHD-GC | ICEA S-75-381 | 2000 V TYPE W | CAM-LOCK | 15 kV MV-105



Type W sets with cam-lock plugs into the 480 V switchboard



WHERE IN THE PLANT

# ACCELERATING TIME TO POWER™

Schedule is decided in the field. The SIMpull Truck delivers multi-conductor SIMpull Reels and pays them off at the pull, so a feeder goes in with one setup instead of three.

**01 SIMPULL REELS**
Paralleled conductors on one reel: one pull per feeder.

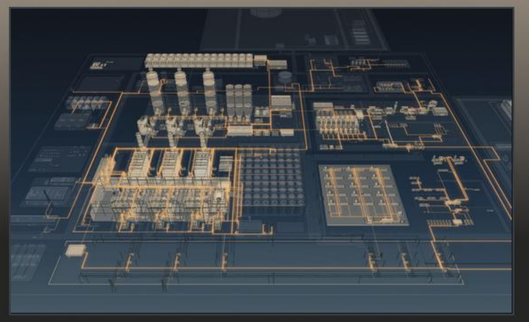
**02 SIMPULL TRUCK**
Delivery, payoff and reel return in one vehicle.

**03 FEWER SETUPS**
Less reel handling on site, faster energization.

SIMPULL® REELS | SIMPULL TRUCK | JOBSITE SERVICES



The cable reel yard: SIMpull Reels staged for the pulls



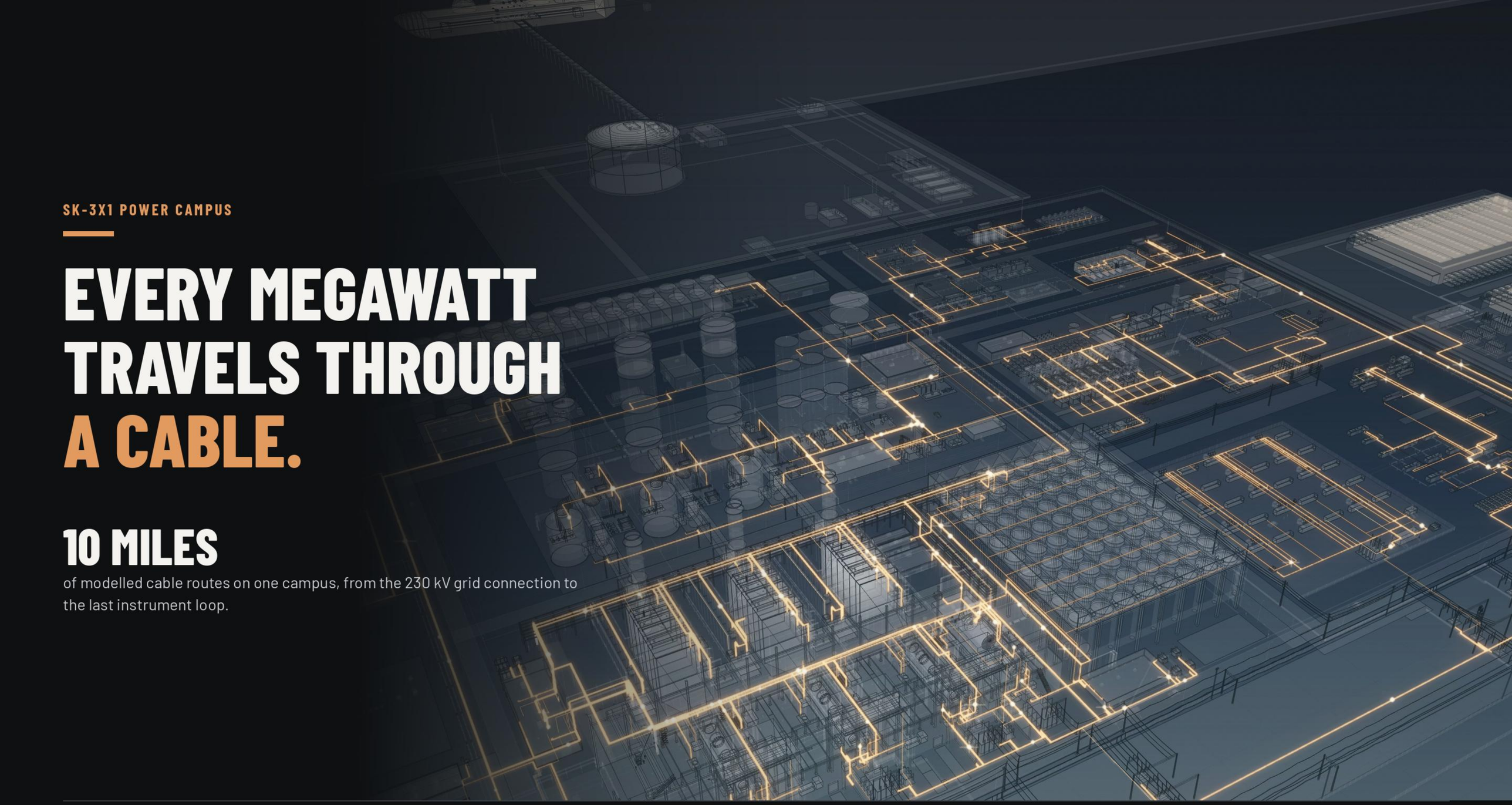
WHERE IN THE PLANT

SK-3X1 POWER CAMPUS

# EVERY MEGAWATT TRAVELS THROUGH A CABLE.

## 10 MILES

of modelled cable routes on one campus, from the 230 kV grid connection to the last instrument loop.



PRODUCT INDEX

# ONE SUPPLIER, GRID TO STACK

| ZONE | PRODUCT | VOLTAGE | STANDARD |
|---|---|---|---|
| GRID CONNECTION | HV XLPE transmission cable | 230 kV | ICEA S-108-720 |
| | Bare copper grounding | 4/0 AWG | — |
| POWER ISLAND | ARMOR-X® MC-HL, red PVC | 15 kV | UL 1569 / 2225 |
| | ARMOR-X® MC-HL, black PVC, CSA RA90-HL | 600 V | UL 1569 / 2225 |
| | MV-105 EPR 133% | 15 kV | UL 1072 |
| | Type TC-ER power and control | 600 V | UL 1277 |
| CARBON CAPTURE | ARMOR-X, MV-105, TC-ER, PLTC | 300 V – 15 kV | UL 13 / 1072 / 1277 |
| BATTERY STORAGE | PV / RHW-2 DC conductor | 2 kV | UL 4703 |
| | MV-105 collection | 35 kV | UL 1072 |
| MODULAR POWER | Type SHD-GC portable power cable | 15 kV | ICEA S-75-381 |
| | Type W portable power cable | 2000 V | UL 1650 |
| INSTALLATION | SIMpull® Reels and SIMpull Truck | — | — |

Southwire®
POWER GENERATION SOLUTIONS

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